

Ian Helfrich, Ph.D.

Applied Economist & Educator · Spatial & Policy Econometrics · Economic Networks · International Trade

Atlanta, GA · ianhelfrich@outlook.com · ianhelfrich.com · github.com/ihelfrich · linkedin.com/in/ian-helfrich

RESEARCH INTERESTS

Applied econometrics for spatial and high-dimensional data, economic networks, international trade, and policy evaluation.

Secondary: regional and environmental economics, remote sensing, optimal transport. *JEL:* C21, C55, C67, F13, F14, R12.

EDUCATION

Ph.D., Economics, Georgia Institute of Technology (*minor: Quantitative Psychology*) 2024

Dissertation: network centrality, spatial heterogeneity, and optimal transport in international trade.

M.S., Geographic Information Science & Technology, Georgia Institute of Technology 2022

M.S., Economics, Georgia Institute of Technology 2019

M.A., Economics, Indiana University 2018

M.Sc., Specialized Economic Analysis: Economics and Public Policy, Barcelona School of Economics 2015

B.A., Economics, University of North Carolina at Chapel Hill 2014

RESEARCH AND WORKING PAPERS

- *AI and the Entry-Level Labor Market*. with S. Vardanyan. Working paper, 2026. Continuous-exposure event-study and triple-difference estimates of generative AI's effect on early-career U.S. employment, using CPS microdata.
- *Trade in the Spotlight*. with E. Gonchar. SSRN 5202676, April 2025. Effective-distance panel framework for bilateral flows under visibility shocks.
- *The Rural Mobilization Gap*. Sole-authored. CDE fixed-effects decomposition of the New Markets Tax Credit; replication at ihelfrich.github.io/us-nmtc-viewer.
- *Metric Sovereignty*. Sole-authored, living draft. Arrow's impossibility, AI governance, and capitalist democracy. ihelfrich.github.io/metric-sovereignty.
- *Penumbra (working title)*. Sole-authored, in progress. Topology and centrality in trade networks.
- *Asymmetric Returns on Exposure*. Theory paper in progress. Networked trade with heterogeneous firm exposure.

Books and Long-Form. *Pictures of Inference: A Visual Approach to Statistics and Econometrics*. Manuscript in preparation, 2026.

Policy Reports. Additional contributor (with E. Gonchar), *Securing America's Future: A Framework for Critical Technology Assessment*, National Network for Critical Technology Assessment (NSF), 2023 — semiconductors skill-gap chapter.

COMPUTATIONAL TOOLS AND OPEN-SOURCE RESEARCH INFRASTRUCTURE

- **ECONSCOPE** – unified economic-intelligence framework (Python/Electron) over the tools below.
- **forensic-econ** (Go) – EDGAR/XBRL filings, Beneish/Altman/Sloan scoring, Wayback, OpenCorporates, OpenSanctions, and CourtListener.
- **deepwire** (Python) – entity-connection graph engine over EDGAR full-text, Wikidata, GDELT, and Reddit using networkx.
- **web-intel** (Go) – parallel crawler producing JSONL plus SHA1-sharded HTML archives (~2,300 URLs/s on sitemap flatten).
- **groundtruth** (Python) – property-records pipeline over NYC GeoSearch/PLUTO/ACRIS and the UK Land Registry for corporate-landlord forensics.
- **company-dissector** (Electron) – 12-tab desktop GUI for company-level dissection across SEC, GDELT, patents, and LittleSis.

Stack. Python (pandas, statsmodels, linearmodels, geopandas, rasterio), R, Stata, Julia, Go; geospatial (WorldPop, GHS-SMOD, ArcGIS Pro, QGIS); LaTeX, Quarto.

APPLIED AND ADVISORY WORK

- **Independent economic advisory**, 2026–present. Industrial-sector diagnostic methodology for a staff member at Saudi Arabia's Ministry of Economy: growth-diagnostic framing, indicator selection and staging, composite-index construction, and international benchmarking design.
- **Quantitative research practice**, 2024–present. Applied econometric and measurement work for institutional and private clients, with a small research lab supporting student-led empirical projects.

TEACHING

- **Instructor of Record**, Georgia Institute of Technology. Three semesters: ECON 200 (twice) and ECON 206.
- **Graduate Teaching Assistant**, Georgia Institute of Technology, 2018–2024. Sections, recitations, and grading across undergraduate economics courses.
- **Graduate Teaching Fellow**, Center for Teaching & Learning (CETL), Georgia Tech, 2018–present. Selective pedagogy-development fellowship.
- **Online and independent teaching**, 2021–present. Roughly 1,200 hours in the past year and 450+ students in econometrics, statistics, and finance (Pepperdine, UC Berkeley, King's College London, Penn State); author of the open-source course companions *stats-lab*, *inference-lab*, *macroprep*, and *econ101a*.

AWARDS, HONORS, AND SERVICE

Graduate Teaching Assistant of the Year, Georgia Institute of Technology 2023

Referee, *Journal of Economic Theory* 2025

VP of Campus Services and Senator, Georgia Tech Graduate Student Government 2018–2020